Business Question and Visualization Report

Date	24 September 2024
Team ID	PNT2022TMIDxxxxxx
Project Name	Health360: Executive Analytics & Population Health Dashboard
Maximum Marks	5 Marks

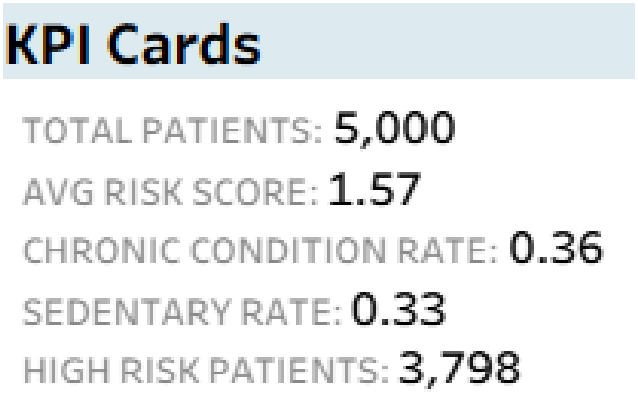
Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

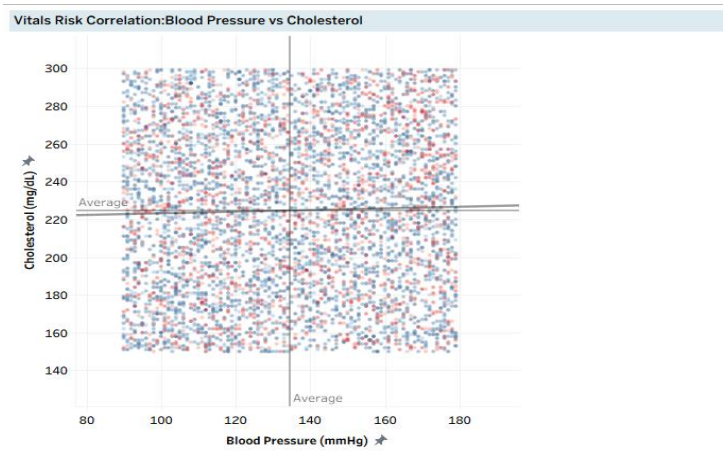
1. Cohort Baseline & Executive KPI Overview

- **Business Questions:**
    1. What is the total patient population size and baseline health risk profile across the entire cohort?
    2. What proportion of the patient population is classified as high risk or exhibits chronic conditions and sedentary lifestyles?
- **Visualization:** Scorecard / Text BANs displaying Total Patients (5,000), Average Risk Score (1.57), Chronic Condition Rate (0.36), Sedentary Rate (0.33), and High-Risk Patient Count (3,798).
- **Chart Type:** Scorecard / Text BANs (Big Angry Numbers)
- **Screenshot**



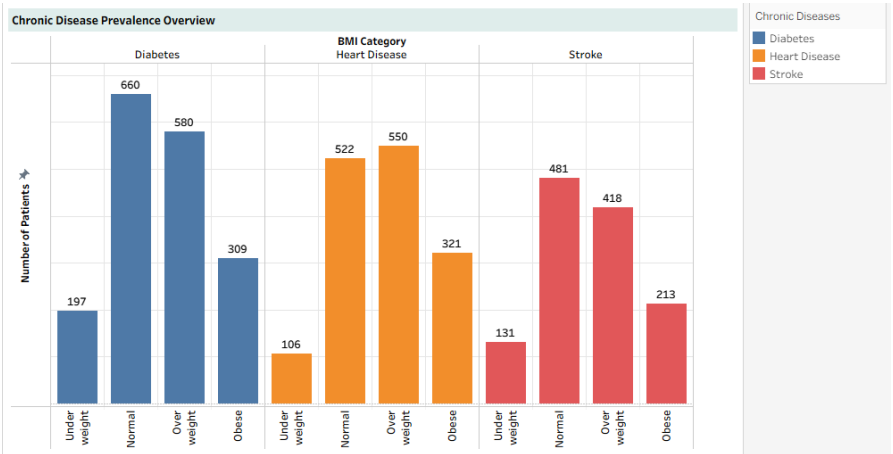
2. Vitals Risk Correlation Analysis (Dr. Michael Carter)

- **Business Questions:**
    1. How do systolic/diastolic blood pressure levels interact with serum cholesterol readings to elevate individual patient risk scores?
    2. Are high-risk patients concentrated in specific blood pressure and cholesterol reference quadrants, or are risks distributed across all vital ranges?
- **Visualization:** Scatter Plot mapping Blood Pressure (mmHg) against Cholesterol (mg/dL) with average reference lines and a colour gradient for Calculated Risk Score (0 to 5).
- **Chart Type:** Scatter Plot (Blood Pressure vs. Cholesterol, colour graded by Calculated Risk Score).
- **Screenshot:**



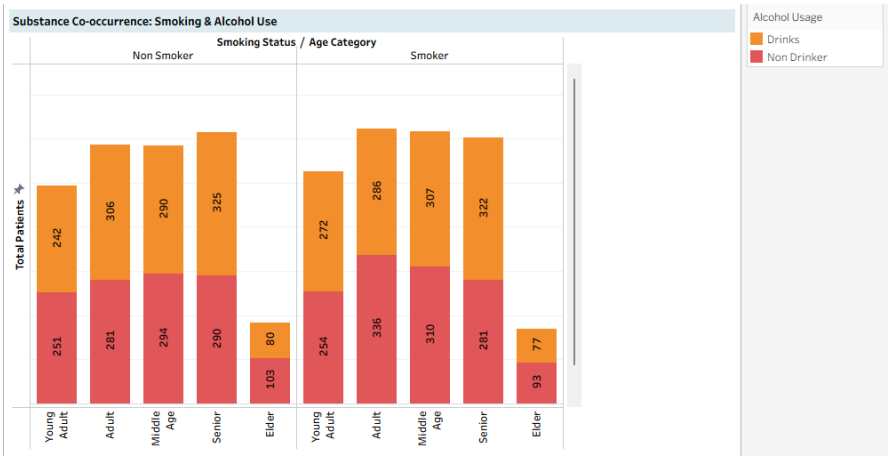
3. Chronic Condition Burden & Co-Occurrence

- **Business Questions:**
    1. How are major chronic conditions (Diabetes, Heart Disease, and Stroke) distributed across different BMI categories (Underweight, Normal, Overweight, Obese)?
    2. Which weight bracket carries the highest total volume of chronic condition cases?
- **Visualization:** Grouped Bar Chart displaying patient counts for Diabetes, Heart Disease, and Stroke segmented by BMI Category.
- **Chart Type:** Grouped Bar Chart
- **Screenshot Description:**



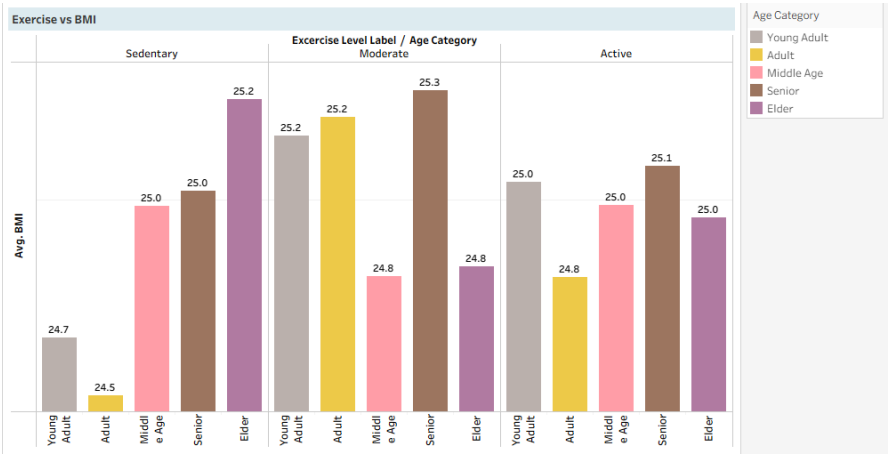
4. Substance Co-occurrence: Smoking & Alcohol Use

- **Business Questions:**
    1. What is the co-occurrence pattern of smoking and alcohol consumption across different patient age categories?
    2. Do non-smokers or smokers exhibit higher rates of alcohol usage across life stages from Young Adults to Elders?
- **Visualization:** Stacked Bar Chart cross-tabulating Smoking Status (Non-Smoker vs. Smoker) and Alcohol Usage (Drinks vs. Non-Drinker) across Age Categories.
- **Chart Types:** Stacked Bar Chart
- **Screenshot:**



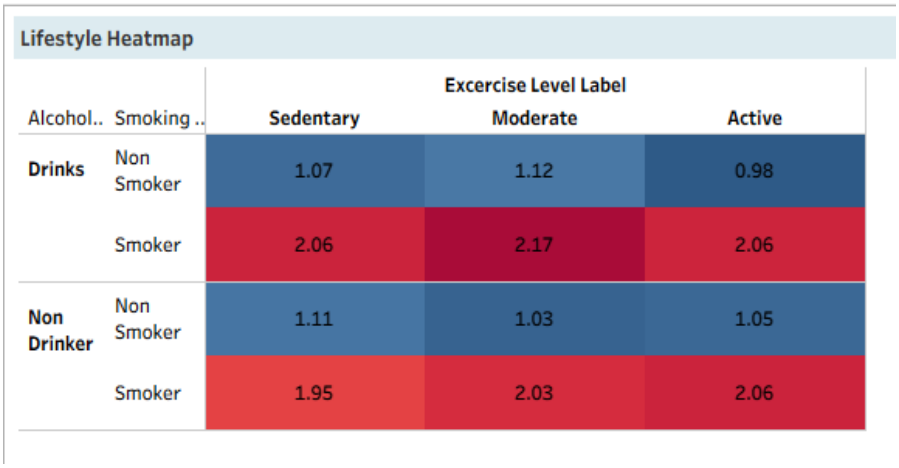
5. Exercise Level vs. Average BMI Analysis

- **Business Questions:**
    1. How does average Body Mass Index (BMI) vary across physical activity levels (Sedentary, Moderate, Active) for each age category?
    2. Do older age brackets consistently maintain higher average BMIs regardless of their physical activity level?
- **Visualization:** Multi-panel Bar Chart comparing Average BMI across Exercise Level Labels (Sedentary, Moderate, Active) broken down by Age Category.
- **Chart Types:** Multi-panel / Grouped Bar Chart
- **Screenshot:**



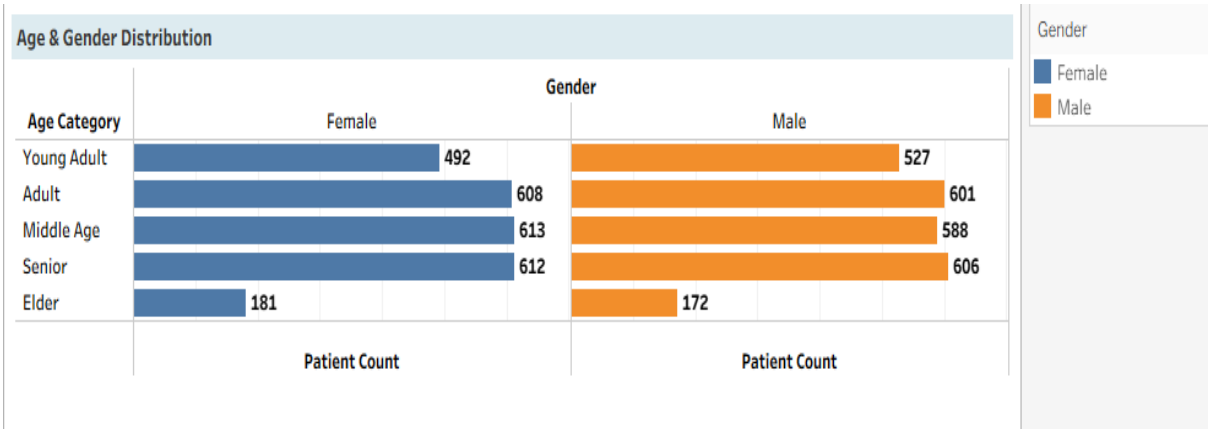
6. Lifestyle Heatmap Matrix

- **Business Questions:**
    1. Which combined lifestyle habit profile (e.g., smoking + drinking + sedentary lifestyle) produces the highest composite risk score?
    2. Does increasing physical activity to 'Active' reduce calculated health risk for patients who drink and smoke?
- **Visualization:** 2-way Heatmap Grid cross-tabulating Alcohol Usage, Smoking Status, and Physical Activity against Average Calculated Risk Scores.
- **Chart Types:** Matrix Heatmap Grid
- **Screenshot:**



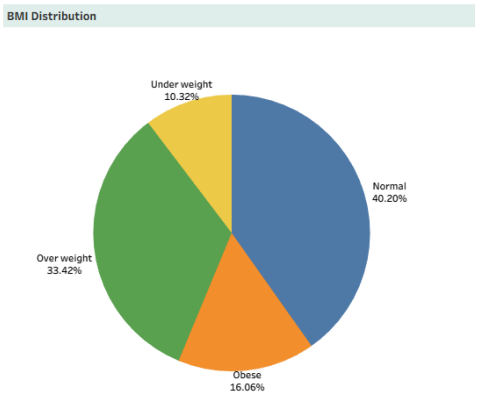
7. Age & Gender Distribution

- **Business Questions:**
    1. How is the 5,000-patient cohort split demographically between male and female populations across life stages?
    2. Which age brackets represent the largest clinical care consumer segments in this healthcare system?
- **Visualization:** Diverging Population Bar Chart / Pyramid showing male and female patient counts across age categories (Young Adult through Elder).
- **Chart types:** Diverging Bar Chart / Population Pyramid
- **Screenshot:**



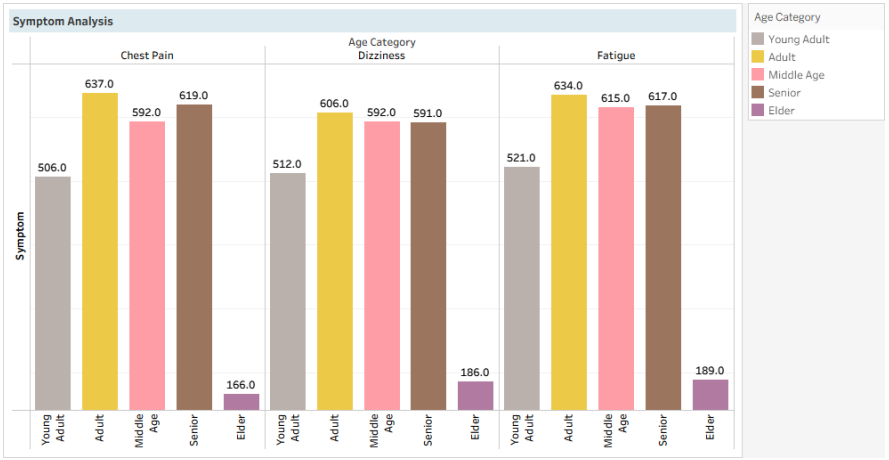
8. BMI Distribution

- **Business Questions:**
    1. What percentage of the overall patient population falls into healthy vs. unhealthy Body Mass Index categories?
    2. What combined proportion of the population is classified as either Overweight or Obese?
- **Visualization:** Pie Chart showing the percentage breakdown across BMI categories (*Normal 40.20%, Overweight 33.42%, Obese 16.06%, Underweight 10.32%*).
- **Chart types:** Pie Chart
- **Screenshot:**



9. Symptom Analysis across Life Stages

- **Business Questions:**
    1. What are the primary reported symptoms (Chest Pain, Dizziness, Fatigue) across different age groups, and which age bracket reports them most frequently?
    2. Are acute symptoms like Chest Pain significantly higher in Adults and Middle-Aged cohorts compared to Elders?
- **Visualization:** Multi-metric Bar Chart comparing total symptom counts (*Chest Pain, Dizziness, Fatigue*) across Age Categories.
- **Chart types:** Multi-metric Bar Chart
- **Screenshot:**



10. Health360 Master Dashboard (Single-Screen View)

- **Business Questions:**
    1. How does selecting a specific patient subgroup (e.g., Obese or Senior) dynamically cross-filter all 9 visual components simultaneously?
    2. How can executive leadership utilize a unified single-screen dashboard to connect clinical vitals, behavioral risk drivers, and population demographics in real time?
- **Visualization:** Health360 Master Dashboard integrating all 9 interactive worksheets into a single executive interface.
- **Chart types:** Multi-metric Bar Chart
- **Screenshot:**

